

Chapter 7

Index Number

Planning operations in various areas, particularly in business management, requires a thorough analysis of the business situation and its characteristics, such as trends in business and changes in different aspects of operations in business. Product prices serve as one of the most critical indicators in business operations, including measurements of purchasing power and numerous other business-related factors. Typically, the rate of change in prices for different products varies considerably. That is, while most product prices may be increasing, some products may experience price decreases, or conversely, while most product prices are declining, certain products may see price increases. In such cases, it becomes evident that analyzing business situations may require multiple indicators, making it difficult to discern trends in general price level changes within business processes. Therefore, it is necessary to identify measurement tools to address this challenge. These tools are designed to help visualize trends in price level changes for either single or multiple products combined. They serve as indicators for business operations. In statistics, the measurement tool designed for this purpose is called an “index number”, which has been widely adopted in the business sector. Index numbers enable clear visualization of important business changes, such as changes in prices of individual products or groups of products, including raw material prices, product prices, production quantities, wage rates, and income – whether monetary income or real income value.

Scope of Study

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7.1 Index Numbers

Index numbers are tools for indicating various situations of interest, enabling us to visualize important changes or to observe movements in data of concern. They may be applied in business or other sectors. For example, in education, educational indices may be used as indicators for educational development. In the business sector, index numbers serve as essential tools for revealing economic and business conditions. They are used to measure the movement and changes of phenomena under study from one time period to another, or from one location to another, using a specific time period or location as a basis for comparison, expressed in the form of ratios or percentages. In general, index numbers are commonly used to provide comparative explanations regarding changes in product prices, quantities, trade values, as well as the values of goods and services, among other applications.

It can be stated that an index number is a value established to serve as a tool for measuring changes in phenomena of interest, primarily used in business to support decision-making in business planning and operations. Index numbers are also used to compare and measure changes in characteristics of interest across various fields between two time periods. In other words, an index number is the ratio of a quantity in one time period compared to the quantity in another time period, commonly expressed as a percentage or a numerical value that demonstrates the relative change of a variable or group of variables from one period to another, compared as a percentage of the base year.

7.2 Construction of Index Numbers

The construction of index numbers should take into account various important characteristics that result in changes or serve as indicators of significant features, capable of guiding various situations. It is necessary to analyze and compare whether the index number is appropriate and how. This may be considered from the following aspects:

1. Purpose of Index Number Construction: to establish clarity in analysis, which will assist in selecting appropriate items that meet requirements, comprehensively cover the subject matter, and ensure accuracy and reliability. Items may be divided into groups, categories, or according to similar characteristics.

2. Data sources used in the analysis must be reliable and appropriately sized according to the type of index number. Additionally, the data collection process must be efficient.

3. Reference Comparison or Base Year Determination: the determination of a base year should consider multiple related factors, such as the purpose of constructing the index number, item selection, and data source determination. A period should be selected when general conditions are normal, that is, the period must not have any critical situations occurring, such as policy changes, new laws, new monetary and fiscal systems, war conditions, severe natural disasters, or other situations that rapidly affect the country's economic condition. The base year must be a period not too distant from the period under study, and if comparative analysis is required, base years should be set at relatively similar time intervals.

4. Selection of Weights are assigned to obtain accurate index values that more closely reflect reality, since different commodities have varying degrees of importance. Commodities of greater importance should be given more weight than others. The selection of weights should align with the purpose of the index number. For example, in constructing a consumer price index, consumer expenditure should be used as the weight. The selection of weights should also consider the method and type of index number. For instance, in calculating price indices, the quantity of goods is used as the weight, while in calculating quantity indices, prices serve as the weight. Furthermore, weights should be periodically updated to reflect current economic conditions, as weights that are appropriate for one period may not be suitable for another period.

7.3 Types of Index Numbers

There are various types of index numbers depending on the subject of interest. For example, if one wishes to study changes in educational

progress, the analyst must identify indicators to reveal how education has advanced. The tools used for such measurement are called educational indices, such as student continuation rates, student dropout rates, and so forth. Similarly, if one is interested in studying population changes, indicators must be identified to demonstrate how the population is changing, such as birth rates, death rates, migration rates, and others. From the examples above, it is evident that index numbers are widely used and serve as excellent tools for examining changes in phenomena of interest. While the index numbers discussed here focus on the business sector, they may be adapted for application in other fields as well. Index numbers may be classified according to the purpose for which they are constructed to measure changes. In economic and business contexts, index numbers can be broadly classified into the following major categories:

1) Price Index is a value used to measure the relative change in prices of goods and services. Price indices are important in economics and business, such as the Wholesale Price Index and the Consumer Price Index.

2) Quantity Index is a value used to measure the relative change in quantities of goods and services. Important quantity indices include, for example, the Export Quantity Index.

3) Value Index is a value used to measure the relative change in the value of goods and services. The value of any commodity is obtained by multiplying the price per unit by the quantity. For example, if the export price of rice is 12,000 baht per ton and 500 tons of rice are exported, then the export value of rice = $12,000 \times 500 = 6,000,000$ baht. It can be observed that the value index reflects changes in both price and quantity simultaneously.

4) Special-Purpose Index Numbers

When classified by construction method, index numbers can be divided into two major categories:

- 1) Unweighted Index
- 2) Weighted Index

7.4 Notation Used in Index Number Calculation

Index number calculations employ various symbols as follows:

p = Price of goods

q = Quantity of goods

v = Value of goods

p_0 = Price of goods in the base period for comparison. For

example, p_{2550} represents the price of goods in 2550, where 2550 is the base year.

q_0 = Quantity of goods in the base period for comparison. For

example, q_{2550} represents the quantity of goods in 2550, where 2550 is the base year.

v_0 = Value of goods in the base period for comparison, where

$$v_0 = p_0 q_0.$$

p_n = Price of goods in the period of interest for which the index number is calculated.

q_n = Quantity of goods in the period of interest for which the index number is calculated.

v_n = Value of goods in the period of interest for which the index number is calculated, where $v_n = p_n q_n$.

I_n = Price index in the period of interest for which the index number is calculated. For example, I_{2552} represents the price index in 2552.

QI_n = Quantity index in the period of interest for which the index number is calculated. For example, QI_{2552} represents the quantity index in 2552.

VI_n = Value index in the period of interest for which the index number is calculated. For example, VI_{2552} represents the value index in 2552.

7.5 Considerations in Constructing Index Numbers

7.5.1 Defining the Purpose of Construction

Since there are various types of index numbers and the purpose of constructing each type differs, it is essential to clearly define the intended purpose before construction. For example, the construction of a consumer price index aims to measure changes in prices of goods and services necessary for living.

7.5.2 Selecting Representative Items of Goods and Services for Index Number Calculation

The accuracy and reliability of an index number depend primarily on a key factor: the selection of groups of goods and services to serve as representatives in calculating the index number, and whether they adequately represent all existing goods and services. For example, in constructing a consumer price index, consumer goods used as representatives in calculating the index number are organized into seven categories, such as food and beverages, housing, medical care, and so forth. Each category contains many types of goods. For instance, when selecting representative items for the food and beverages category, the selector must possess expertise in determining which food and beverage prices will serve as the best representatives.

7.5.3 Selecting the Base Period for Comparison

The selection of a base period may be a year, month, or week. However, the base period should be a normal period, such as one without war conditions or with overall normal economic conditions. When comparing the period of interest to observe changes, the time intervals should be equal.

7.5.4 Selecting Formulas and Weighting Methods

The same type of index number may have multiple calculation formulas and different weighting methods. However, the selection of

formulas and weighting methods are interrelated. The choice of which formula to use depends on the constructor's judgment regarding which formula is most appropriate. The construction of each type of index number includes both unweighted and weighted calculation methods.

Each type of index can be calculated from either a single item or multiple items. When calculated from a single item, it is called a simple index, such as a simple price index or simple quantity index. When calculated from multiple items, it is called a composite index. If classified by calculation method, each type of index can be divided into two categories: unweighted index and weighted index. The detailed calculations for these indices will be discussed in subsequent sections.

7.6 Price index

The calculation of a price index is based on the principle of comparing the price of goods in the year of interest with the price of goods in the base year. The goods for which the price index is calculated may be a single item or multiple items combined.

An unweighted price index is calculated by giving equal importance to all items. Unweighted price indices are divided into three types: simple price index, simple aggregative price index, and simple average of relative price index, as follows:

7.6.1 Simple price index

A simple price index is a value that describes the change in price of a single item in the year of interest compared to the price in the base year, expressing the change as a relative price. The calculation formula is as follows:

Simple Price Index $(I_n) = \frac{P_n}{P_0} \times 100$

Example 7.1 Suppose the price of a computer from a certain company in 2550 was 38,600 baht, and in 2551 the price was 32,000 baht. Find the simple price index for 2551 compared to 2550.

Solution:

1) The base year is 2550.

2) Calculate the simple price index as follows: $I_n = \frac{P_n}{P_0} \times 100$

$$\begin{aligned}
 I_{2551} &= \frac{P_{2551}}{P_{2550}} \times 100 \\
 &= \frac{32,000}{38,600} \times 100 \\
 &= 82.9
 \end{aligned}$$

3) This indicates that the computer price in 2551 decreased from the 2550 price by 17.1%.

7.6.2 Simple aggregative price index

A simple aggregative price index is a value that describes the change in prices of a group of goods in the year of interest compared to the prices of that group of goods in the base year. The calculation formula is as follows:

$$\text{Simple Aggregative Price Index } (I_n) = \frac{\sum p_n}{\sum p_0} \times 100$$

Example 7.2 Suppose that the prices of five consumer goods in the years 2551-2552 are shown in the table below. Find the simple aggregative price index for the year 2552 relative to 2551.

Product list	Product price (Baht/Unit)	
	2551	2552
Rice (kilogram)	15	18
Pork (kilogram)	90	95
Eggs size 0 (dozen)	35	40
Cooking oil (liter)	56	62
Sugar (kilogram)	20	22
Total	216	237

Solution:

- 1) The base year is 2551.
- 2) Calculate the simple aggregative price index as follows:

$$I_n = \frac{\sum p_n}{\sum p_0} \times 100$$

$$\begin{aligned}
 I_{2552} &= \frac{\sum p_{2552}}{\sum p_{2551}} \times 100 \\
 &= \frac{237}{216} \times 100 \\
 &= 109.72
 \end{aligned}$$

3) This result indicates that the overall prices of the five consumer goods in 2552 increased from 2551 by 9.72%.

7.6.3 Simple average of relative price index

A simple average of relative price index is a value that describes the change in prices of a group of goods in the year of interest compared to the prices of that group of goods in the base year, similar to the simple aggregative price index. However, the relative price of each individual item is calculated first, then the relative prices of all items are summed together and averaged by dividing by the number of items. This calculation method eliminates the problem of different price magnitudes among goods, making the observed change a true reflection of relative price changes. Therefore, it is more accurate and reliable than the simple aggregative price index. The calculation formula is as follows:

$\text{Simple Average of Relative Price Index } (I_n) = \frac{\sum \frac{p_n}{p_0}}{n} \times 100$
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Example 7.3 Using the data from Example 7.2, find the simple average of the relative price index.

Solution:

1) The base year is 2551.

2) Calculate the simple average of the relative price index as

follows: $I_n = \frac{\sum \frac{p_n}{p_0}}{n} \times 100$ therefore $I_{2552} = \frac{\sum \frac{p_{2552}}{p_{2551}}}{5} \times 100$

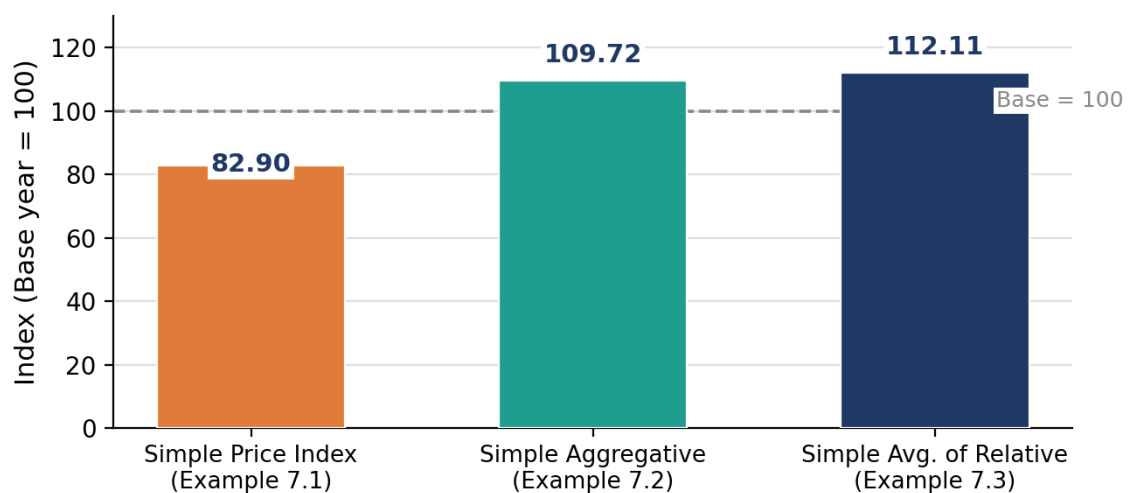
3) Calculate the various values as follows:

Product list	Product price (Baht/Unit)		$\frac{p_{2552}}{p_{2551}}$
	2551	2552	
Rice (kilogram)	15	18	$\frac{18}{15} = 1.20$
Pork (kilogram)	90	95	$\frac{95}{90} = 1.06$
Eggs size 0 (dozen)	35	40	$\frac{40}{35} = 1.14$
Cooking oil (liter)	56	62	$\frac{62}{56} = 1.11$
Sugar (kilogram)	20	22	$\frac{22}{20} = 1.10$
Total	216	237	5.61

4) Calculate the value $I_{2552} = \frac{5.61}{5} \times 100 = 112.11$

5) This indicates that the prices of these 5 consumer goods in 2552 increased from 2551 by 12.11%.

Comparing the Three Unweighted Price Index Methods



Note: Example 7.1 uses a single product (computer price); Examples 7.2-7.3 use the same 5-item consumer goods data set, comparing prices in 2552 to base year 2551.

Figure 7.1 Comparison of the three unweighted price index methods (Examples 7.1-7.3)

7.7 Quantity index

A quantity index is a value that indicates the change in quantities of goods and services in the year of interest compared to the base year. The quantity index is used to measure this change and is particularly useful in industry, as changes in product quantities affect production planning. The construction of quantity indices follows principles similar to price indices, except that price is replaced by quantity.

An unweighted quantity index follows the same principle as an unweighted price index, which is calculating the index by giving equal importance to all items. Unweighted quantity indices are divided into three types:

7.7.1 Simple quantity index

A simple quantity index is a value that describes the change in the quantity of a single item in the year of interest compared to the quantity in the base year. The calculation formula is as follows:

Simple Quantity Index $(QI_n) = \frac{q_n}{q_0} \times 100$
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7.7.2 Simple aggregative quantity index

A simple aggregative quantity index is a value that describes the change in quantities of a group of goods in the year of interest compared to the quantities of that group of goods in the base year. The calculation formula is as follows:

Simple Aggregative Quantity Index $(QI_n) = \frac{\sum q_n}{\sum q_0} \times 100$
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7.7.3 Simple average of relative quantity index

A simple average of relative quantity index is a value that describes the change in quantities of a group of goods in the year of interest compared to the quantities of that group of goods in the base year. The relative quantity of each individual item is calculated first, then the relative quantities of all items are summed together and averaged by dividing by the number of items. The calculation formula is as follows:

Simple Average of Relative Quantity Index

$$(QI_n) = \frac{\sum \frac{q_n}{q_0}}{n} \times 100$$

Example 7.4 The data in the table below shows the employment numbers for each of 4 sectors over a 2-year period, from 2550 to 2551, as follows:

Sector	Employment Quantity (persons)	
	2550	2551
Agriculture	16,031	15,945
Manufacturing	4,357	4,178
Industry	44	40
Mining	2,021	1,763
Construction		
Total	22,453	21,926

- Calculate the simple aggregative quantity index for employment in 2551 using 2550 as the base year.
- Calculate the simple average of relative quantity index for employment in 2551 using 2550 as the base year.

Solution:

a) Since 2550 is used as the base year, the simple aggregative quantity index is calculated as follows:

$$\begin{aligned}
 QI_n &= \frac{\sum q_n}{\sum q_0} \times 100 \\
 QI_{2551} &= \frac{\sum q_{2551}}{\sum q_{2550}} \times 100 \\
 &= \frac{21,926}{22,453} \times 100 \\
 &= 97.65
 \end{aligned}$$

This indicates that total employment in 2551 decreased from 2550 by 2.35%.

b) Since 2550 is used as the base year, the simple average of relative quantity index is calculated as follows:

$$QI_n = \frac{\sum \frac{q_n}{q_0}}{n} \times 100$$

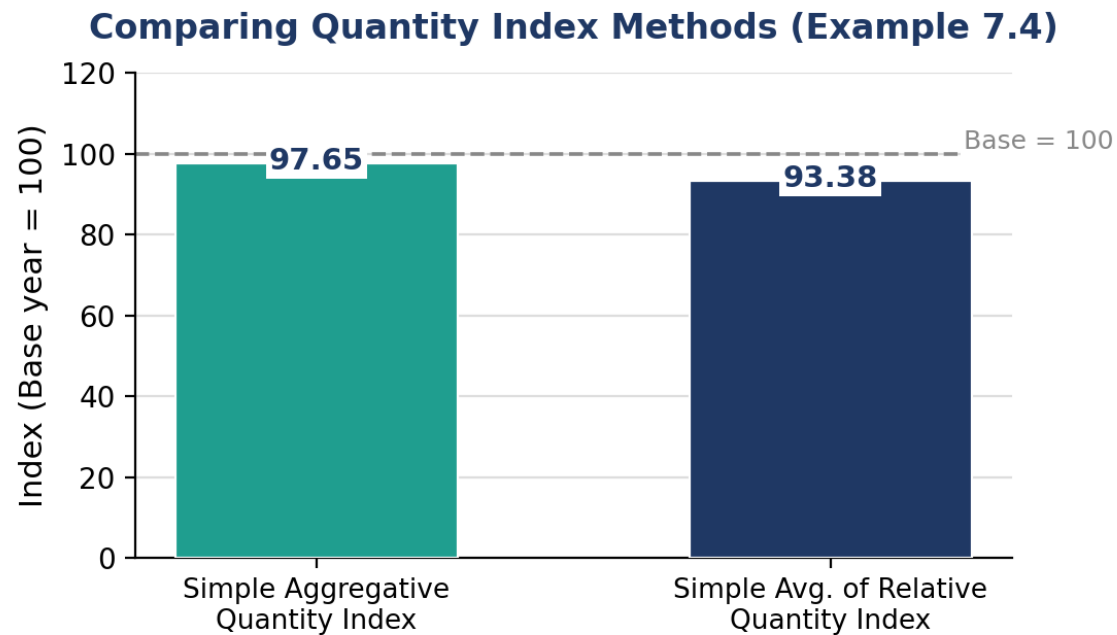
$$QI_{2551} = \frac{\sum \frac{q_{2551}}{q_{2550}}}{4} \times 100$$

Calculate the various values as follows:

Sector	Employment Quantity (persons)		$\frac{q_{2551}}{q_{2550}}$
	2550	2551	
Agriculture	16,031	15,945	$\frac{15,945}{16,031} = 0.995$
Manufacturing Industry	4,357	4,178	$\frac{4,178}{4,357} = 0.959$
Mining	44	40	$\frac{40}{44} = 0.909$
Construction	2,021	1,763	$\frac{1,763}{2,021} = 0.872$
Total	22,453	21,926	3.735

$$QI_{2551} = \frac{3.735}{4} \times 100 = 93.38$$

This indicates that total employment in 2551 decreased from 2550 by 6.62%.



Note: Based on employment data for 4 sectors, comparing 2551 to base year 2550.

Figure 7.2 Comparison of quantity index methods (Example 7.4)

7.8 Value index

A value index is a value that measures the change in the value of goods and services from one period to another, such as the change in sales value of a certain product from 2551 to 2552. The calculation of a value index uses the same principle as that used in calculating a price

index. From the value of goods $(v) = pq$ and the total value of goods

$\sum v = \sum pq$, the simple aggregative value index is the relative ratio that shows the change in total value of goods in the year of interest compared to the base year. The calculation formula is summarized as follows:

Simple Aggregative Value Index	$(VI_n) = \frac{\sum p_n q_n}{\sum p_0 q_0} \times 100$
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In cases where it is desired to calculate a weighted aggregate value index, it is called a weighted average of relative value index, with the following calculation formula:

Weighted Average of Relative Value Index	
$(VI_n) = \frac{\left(\sum \frac{p_n q_n}{p_0 q_0} \times v \right)}{\sum v} \times 100$	
by	$v = \begin{cases} p_0 q_0 & \text{when using base year value as weight} \\ p_n q_n & \text{when using current year value as weight} \\ p_a q_a & \text{when using any year's value as weight} \end{cases}$

Example 7.5 The table below shows data on pharmaceutical sales for disease prevention and treatment for 2549 and 2551. Calculate the weighted average of the relative value index when the base year value of 2549 is designated as the weight.

Medicine	Price (Baht)		Quantity (tablets)	
	2549	2551	2549	2551
Penicillin	5	9	25,000	29,500
Streptomycin	10	12	18,000	24,000
Alkaloid	8	13	13,400	17,000

Solution:

1) Since the base year value of 2549 is designated as the weight,

$$v = v_0 = p_0 q_0 = p_{2549} q_{2549}$$

2) Calculate the weighted average of relative value index as follows:

$$VI_n = \frac{\left(\sum \frac{p_n q_n}{p_0 q_0} \times v \right)}{\sum v} \times 100$$

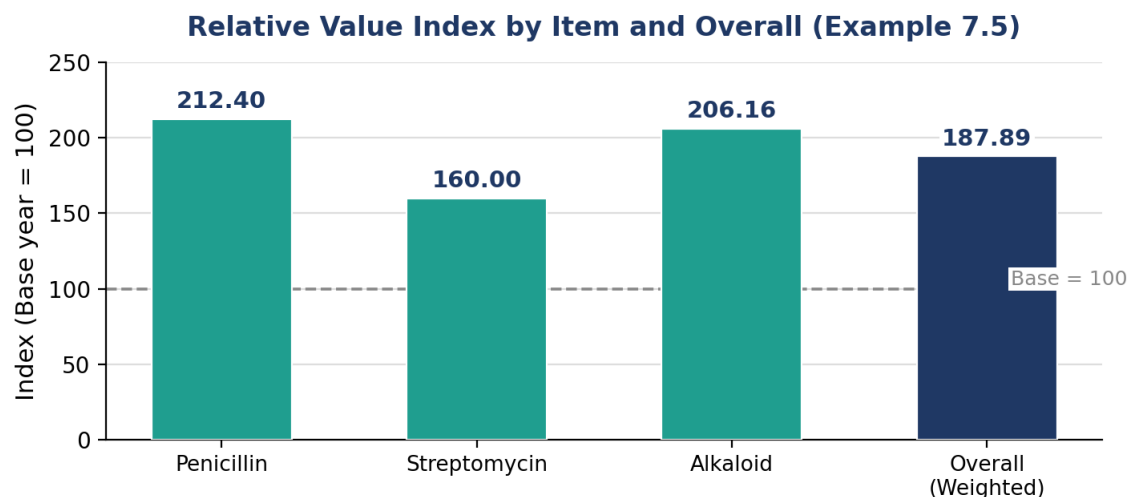
$$VI_{2551} = \frac{\left(\sum \frac{p_{2551} q_{2551}}{p_{2549} q_{2549}} \times p_{2549} q_{2549} \right)}{\sum p_{2549} q_{2549}} \times 100$$

3) Calculate the various values as follows:

Medicine	$p_{2551}q_{2551}$	$p_{2549}q_{2549}$	$\sum \frac{p_{2551}q_{2551}}{p_{2549}q_{2549}} \times p_{2549}q_{2549}$
Penicillin	265,500	125,000	$\frac{265,500}{125,000} \times 125,000 = 265,500$
Streptomycin	288,000	180,000	$\frac{288,000}{180,000} \times 180,000 = 288,000$
Alkaloid	221,000	107,200	$\frac{221,000}{107,200} \times 107,200 = 221,000$
Total		412,200	774,500

4) Therefore, $VI_{2551} = \frac{774,500}{412,200} \times 100 = 187.89$

5) This indicates that the sales value of pharmaceuticals in 2551 increased from 2549 by 87.89%.



Note: Individual relative value index per medicine ($p_1q_1/p_0q_0 \times 100$), 2549 base vs 2551, compared with the overall weighted average of relative value index.

Figure 7.3 Relative value index by item and overall weighted index (Example 7.5)

7.9 Consumer Price Index: CPI

The consumer price index, or cost of living index, is a tool used to measure changes in the prices of goods and services that most consumers regularly purchase in fixed quantities and quality as defined

in the base year. Consumers refer to households with regular income or salaries, such as employees and civil servants.

The goods and services used to calculate the consumer price index are divided into 7 categories as follows:

1) Food and Beverages: this includes rice, flour, flour products, meat, duck, chicken, aquatic animals, vegetables, fruits, eggs, milk, dairy products, food purchased from markets, and non-alcoholic beverages.

2) Housing: this includes accommodation costs, house rent, construction costs, household furnishings, electricity, water supply, fuel, household textiles, and household cleaning materials.

3) Clothing and Footwear: this includes clothing for men and boys, clothing for women and girls, and tailoring services.

4) Medical Care and Personal Services: this includes medical examination fees, medicine costs, and personal service fees.

5) Transportation and Communication

6) Recreation, Reading, and Education

7) Tobacco and Alcoholic Beverages

These 7 categories of goods and services are items that most households in the country regularly purchase. The Economic Index Division of the Department of Trade and Economic Indices, Ministry of Commerce, is responsible for collecting price data on various goods and services to calculate the consumer price index and publish changes in the prices of these goods and services. This serves as an indicator of changes in general economic conditions.

The consumer price index indicates the purchasing power of consumers, showing whether it has increased or decreased and by how much. Therefore, it is an important tool for businesses or the government to use as a guideline in adjusting salaries, wages, minimum wages, rent, or other social assistance. Additionally, the consumer price index is used in calculating the purchasing power of money.

The determination of weights requires knowledge of the quantities of goods purchased for consumption in the base year. The quantities of goods regularly consumed are determined from households that represent general consumers in various areas, including municipal

zones in provinces across Thailand's 4 regions, in the Bangkok metropolitan area, and in surrounding areas.

The selection of product items is derived from choosing goods and services from the 7 categories. Price collection for calculation purposes is conducted in both Bangkok and provincial areas. Retail prices are collected from markets in Bangkok, markets in surrounding areas, and markets in provincial regions across all 4 regions of the country.

Benefits of the Consumer Price Index

The consumer price index is used to measure the cost of living as well as monetary conditions, such as measuring the inflation rate. The inflation rate is the rate of increase in commodity prices. For example, if the consumer price index in 2538 equals 154 (when the base year is 2529), this indicates that over the 9-year period from 2529 to 2538, the inflation rate was 54 percent, or an average inflation rate of approximately 6 percent per year. Additionally, it is used to measure the purchasing power of money or the real value of money. The purchasing power of money indicates what one unit of currency is actually worth in comparison to the economic conditions at that time, or it is used to measure real income. The important formulas related to the application of the consumer price index are as follows:

$$\text{Purchasing Power of Money} = \frac{1}{\text{Consumer Price Index}} \times 100$$

$$\text{Real Income} = \frac{\text{Current Monetary Income}}{\text{Consumer Price Index}} \times 100$$

Furthermore, the consumer price index is used as a deflator (statistical deflation). Since value is the product of price and quantity, value may change due to price alone or due to quantity alone. Therefore, values can be adjusted according to the desired price level as follows:

$$\text{Value at Desired Price Level} = \frac{\text{Current price value}}{\text{Consumer Price Index}} \times 100$$

Example 7.6 If the consumer price index and monthly income of Mr. Tawan from 2548-2552 are as shown in this table:

Year	2548	2549	2550	2551	2552
Income (Baht)	10,200	10,700	11,600	12,800	13,700
Consumer Price Index	100	105	110.2	114.8	120.5

- Calculate the purchasing power for each year when 2548 is designated as the base year.
- Find Mr. Tawan's real income for each year when 2548 is designated as the base year.
- What should Mr. Tawan's monthly income be in 2552 to maintain the same purchasing power as in 2548?
- If Mr. Tawan purchased a house in 2548 for 900,000 baht and wants to sell the house in 2552, at what price should he sell it to maintain the same value?

Solution:

- From the formula:

$$\text{Purchasing Power of Money} = \frac{1}{\text{Consumer Price Index}} \times 100$$

Calculate the purchasing power for each year as follows:

Year	Consumer Price Index (Base Year 2548)	Purchasing Power of Money
2548	100	$\frac{1}{100} \times 100 = 1$
2549	105	$\frac{1}{105} \times 100 = 0.95$
2550	110.2	$\frac{1}{110.2} \times 100 = 0.91$
2551	114.8	$\frac{1}{114.8} \times 100 = 0.87$
2552	120.5	$\frac{1}{120.5} \times 100 = 0.83$

The calculated purchasing power can be interpreted as follows: In 2548, the purchasing power equals 1, meaning 1 baht has a purchasing power equal to 1 baht. In 2549, the purchasing power equals

0.95, meaning 1 baht has the purchasing power equivalent to 0.95 baht when compared to 2548, and so on.

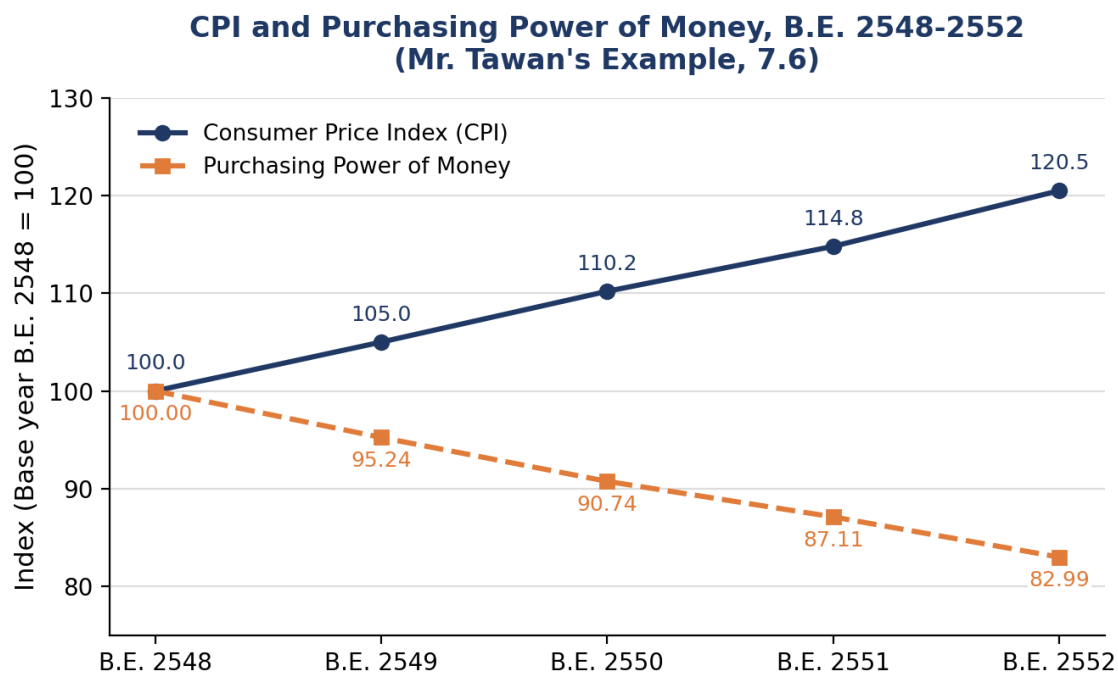


Figure 7.4 CPI and purchasing power of money, B.E. 2548-2552 (Example 7.6)

b) From the formula:

$$\text{Real Income} = \frac{\text{Current Monetary Income}}{\text{Consumer Price Index}} \times 100$$

Calculate the real income as follows:

Year	Consumer Price Index (Base Year 2548)	Nominal Income (Baht)	Real Income (Baht)
2548	100	10,200	$\frac{10,200}{100} \times 100 = 10,200$
2549	105	10,700	$\frac{10,700}{105} \times 100 = 10,190.48$
2550	110.2	11,600	$\frac{11,600}{110.2} \times 100 = 10,526.32$
2551	114.8	12,800	$\frac{12,800}{114.8} \times 100 = 11,149.83$
2552	120.5	13,700	$\frac{13,700}{120.5} \times 100 = 11,369.29$

The calculated real income can be interpreted as follows: In 2548, the real income equaled 10,200 baht, which equals the nominal income because 2548 is used as the base year. In 2549, the real income equaled 10,190.48 baht. It can be observed that although the nominal income in 2549 increased from 2548 by $10,700 - 10,200 = 500$ baht, the rate of income increase was less than the rate of increase in consumer goods prices. This resulted in real income actually decreasing by $10,200 - 10,190.48 = 9.52$ baht. The real income for other years can be explained in a similar manner.

c) Let S represent the monthly income in 2552.

Purchasing Power of Money in 2548 = Purchasing Power of Money in 2552

$$\begin{aligned}\frac{1}{100} \times 100 \times 10,200 &= \frac{1}{120.5} \times 100 \times S \\ S &= \frac{120.5}{100} \times \frac{100}{100} \times 10,200 \\ &= \frac{120.5}{100} \times 10,200 \\ &= 12,291 \text{ baht}\end{aligned}$$

Therefore, Mr. Tawan should have a monthly income of 12,291 baht in order to maintain the same purchasing power as in 2548.

d) The consumer price index for 2552 compared to 2548 equals 120.5. From the formula:

$$\begin{aligned}\text{Value at Desired Price Level} &= \frac{\text{Current price value}}{\text{Consumer Price Index}} \times 100 \\ 900,000 &= \frac{\text{Current price value}}{120.5} \times 100\end{aligned}$$

Therefore, the house should be sold at $= \frac{900,000 \times 120.5}{100} = 1,084,500$ baht